

Algorithms and Responsibility

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Abstract

A long line of experimental work suggests that decision-makers can—and do—use delegation to shift responsibility for unwelcome outcomes onto an intermediary. We ask whether the same logic carries over to delegation to algorithms. In a preregistered online experiment ($N = 322$ UK Prolific participants, 161 in each role), a decision-maker makes a payoff-relevant prediction for a matched recipient with the option to delegate to an algorithm calibrated, at the individual level, to perform as well as she does. We manipulate, between subjects, whether the matched recipient can punish her for the decision. Two findings emerge. First, contrary to the preregistered prediction of the responsibility-avoidance literature, the *prospect* of punishment *reduces* rather than increases the likelihood of delegating to the algorithm (a 17 percentage-point gap, two-sided $p = 0.033$). Second, conditional on outcome, we cannot reject equal punishment of delegated and self-made decisions; the point estimate runs in the predicted direction but the test is underpowered to rule out moderate insulation effects. We discuss several candidate mechanisms for the headline reversal and argue that, among those we can evaluate, a process-ownership account—in which the moral salience introduced by the punishment possibility raises the value of being the proximate cause of a good outcome—is the most consistent with the data, while flagging that the supporting evidence is exploratory and the design cannot rule out experimenter demand. Within the bounds of the setting we study, our results offer a behavioral complement to regulatory frameworks that hold human users accountable for algorithm-mediated decisions: such frameworks may discipline algorithmic adoption rather than merely allocate post-hoc responsibility.

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1 Introduction

Algorithms are increasingly involved in consequential decisions: they screen job candidates, set credit scores, recommend medical treatments, evaluate parole applicants, and—more mundanely but at vast scale—route the cars driving toward this very page. As the locus of decision-making shifts toward algorithmic systems, a parallel question has moved into public, legal, and academic debate: when an algorithm-mediated decision turns out badly, who bears the blame? A natural intuition, which finds support in a long line of experimental work on responsibility avoidance through delegation, is that decision-makers can use algorithms as a means to shift blame onto a non-human intermediary. To return to a familiar example: if you and a friend arrive late because she relied on a navigation app rather than her own sense of direction, the algorithm seems a convenient scapegoat—and the next time she is unsure of the route, the prospect of avoiding your blame may itself motivate her use of the app. This paper provides experimental evidence on whether either of these intuitions is borne out.

We address two questions. *First*: can decision-makers actually shift responsibility away from themselves by delegating to an algorithm—that is, are they punished less for a delegated decision than for a self-made decision with the same outcome? *Second*: does the prospect of being held responsible motivate the decision to delegate in the first place? The two questions correspond to, respectively, the *viability* and the *motivational salience* of algorithmic responsibility avoidance. They have been studied in different settings and rarely in combination, and previous designs have struggled to identify them cleanly. We do so here in a preregistered online experiment.

Our experimental design pins down both questions. Player A first completes ten rounds of an incentivised visual prediction task (estimating a person’s weight from a facial image and stated height). She then makes an final prediction whose outcome determines the payoff of a second participant, Player B: a high payoff if the prediction falls within ten pounds of the true weight, a low payoff otherwise. Player A may either make this prediction herself or delegate it to an algorithm. The algorithm is calibrated to mimic Player A’s individual performance from the first ten rounds—that is, on the final prediction it produces the high outcome with the same probability as Player A herself. Player A is told this. We manipulate the punishment possibility between subjects: in the *Punishment* condition, Player B may reduce Player A’s payoff conditional on the delegation decision and the realized outcome; in the *No-Punishment* condition, no punishment is possible. Player B’s punishment behavior is elicited via the strategy method across all four (delegation, outcome) cells, providing a within-subject test of whether delegation insulates Player A from punishment.

Two design features distinguish our setting from the closest precedent, [Feier et al. \(2022\)](#). First, the algorithm is calibrated at the *individual* level rather than the session or group level. Existing designs typically expose subjects to a distribution of algorithmic performance and let them infer their own ability relative to that distribution; the delegation decision then carries strong signal value about the principal’s self-perception, and any differential punishment can reflect

overconfidence rather than responsibility shifting per se. By neutralizing relative-performance signaling at the individual level, we isolate the responsibility-shifting motive. Second, our *No-Punishment* condition turns off the punishment possibility entirely, allowing us to identify the effect of anticipated punishment on the delegation decision itself—a test that is not feasible in designs where punishment is always possible. Both features sharpen the test of responsibility shifting in algorithmic delegation that the literature has thus far been able to deliver.

Two findings emerge, and in their joint structure they push against the natural reading of the responsibility-avoidance literature. *First, the prospect of punishment reduces, rather than increases, the likelihood of delegation:* 42.5% of Player As delegate when punishment is possible, compared to 59.3% when it is not (a 17 percentage-point gap; two-sided Pearson Chi-squared $p = 0.033$, Fisher exact $p = 0.041$; the preregistered one-sided test in the H1 direction is non-significant by construction, since the result reverses the predicted direction—we discuss this in detail in Section 4). This rejection of the standard prediction in favor of the opposite direction is the paper’s main empirical contribution, and we view it as a behavioral counterpart to the regulatory intuition that holding humans accountable for algorithm-mediated decisions disciplines their use of those algorithms. *Second, conditional on outcome, we cannot reject equal punishment of delegated and self-made decisions in our setting:* Player B’s punishment of Player A is statistically indistinguishable across the two, and Player A correctly anticipates this absence of differential punishment in her elicited beliefs. The point estimate of any insulation effect runs in the predicted direction but the test is underpowered against moderate alternatives ($n \approx 67$ Player Bs after attention-check exclusion); the broader claim that responsibility shifting through algorithmic delegation is altogether absent is one we cannot make with this data.

What mechanism reconciles the two findings? Among the candidates we can evaluate with our design, a *process-ownership* account is the most consistent with the data—though we treat it as a candidate hypothesis rather than an established mechanism, since the supporting evidence is exploratory and a fourth candidate (experimenter demand) cannot be ruled out by construction. A simple anticipation story—in which Player A delegates less when punishment is on the table because she expects harsher punishment for delegated decisions—is inconsistent with the elicited beliefs and with Player B’s actual choices. A simple effort story—in which Player A makes the prediction herself in order to outperform the algorithm and earn the high payoff—is inconsistent with the data, in which non-delegators in the *No-Punishment* condition outperform non-delegators in the *Punishment*. What remains is the milder hypothesis that, when the moral salience of the decision is heightened by the punishment possibility, Player A wants to be the proximate cause of any good outcome for Player B, irrespective of any quantifiable belief about how she will be judged. Consistent with this interpretation, the negative correlation between Player A’s task performance and her propensity to delegate is observed in the *Punishment* condition only: those most likely to deliver the high payoff want to do so personally when the moral stakes are made salient. Section 4 weighs the four candidate mechanisms in detail and is explicit about what the data can and cannot rule out.

These findings contribute to three strands of literature. First, we add to the experimental literature on responsibility avoidance through delegation (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013; Hamman et al., 2010; Hill, 2015; Steffel et al., 2016), by extending the canonical setting to delegation to an algorithmic intermediary in a setting with genuine outcome uncertainty. Second, we contribute to the rapidly growing literature on algorithm aversion and adoption (Dietvorst et al., 2015; Logg et al., 2019; Burton et al., 2020; Mahmud et al., 2022; Chugunova and Sele, 2022) by isolating an under-studied determinant of algorithm use: the institutional context in which the human user is held accountable. Third, and most directly, we build on Feier et al. (2022) and Kirchkamp and Strobel (2019) by providing a clean experimental test of two long-standing claims in the responsibility-and-algorithms literature, with a design that controls relative-performance signaling at the individual level and that includes a condition in which punishment is ruled out by construction. The contrast with Feier et al. (2022) is consequential: that study finds that decision-makers are held *less* responsible for algorithmically delegated decisions, while we find that the punishment possibility *reduces* delegation rather than increasing it. We argue in Section 2 that the two findings are reconcilable: the relative-performance signaling that drives much of Feier et al.’s differential punishment is closed off by construction in our design, isolating a different and milder margin (the motivational salience of accountability for the principal). Beyond the experimental literature, our findings speak to the regulatory debate over the assignment of responsibility for algorithmic decisions (European Union, 2024): within the bounds of the setting we study, holding human users accountable for algorithm-mediated outcomes appears to be a behavioral lever for direct human engagement with the decision, rather than merely a tool for ex-post moral attribution. We are explicit in Section 5 about why the present evidence—a single online experiment on a stylised prediction task with one UK Prolific sample—supports this claim only in a bounded sense, and what would be required to extend it to the high-stakes algorithmic domains (medical, judicial, hiring, financial) that motivate the regulatory debate.

The remainder of the paper is organized as follows. Section 2 reviews the relevant literature in detail. Section 3 describes the experimental design and states our two preregistered hypotheses. Section 4 presents the empirical findings, including the mechanism analysis. Section 5 discusses the implications and limitations of our results.

2 Related Literature

This paper bridges three strands of literature: (i) responsibility avoidance through delegation, an established line of experimental work in which the intermediary is another human; (ii) algorithm aversion and adoption, which examines when and why people are willing to use algorithmic decision aids; and (iii) responsibility attributions to decisions involving algorithms, an emerging literature at the intersection of the first two. We review each in turn, draw on a fourth body of work—identity, moral self-image, and diffusion of responsibility—to motivate

our preferred mechanism, and conclude by clarifying the gap our experiment addresses.

Responsibility avoidance through delegation

The political-science and public-choice literatures have long recognized that delegation can serve as a strategy for shifting blame onto an intermediary (Weaver, 1986; Fiorina, 1986). Experimental economics has established the same point in incentivised settings, almost always using modified dictator games.

Bartling and Fischbacher (2012) provide the canonical experimental demonstration. In their design, a dictator may either choose between a fair and an unfair allocation of an endowment or instead delegate this choice to an intermediary whose monetary incentives are aligned with the dictator's. The recipient is adversely affected if the unfair allocation is chosen and may punish the dictator and the intermediary. Punishment is effectively shifted from the dictator to the intermediary, and the prospect of avoiding punishment constitutes a strong motive for the delegation decision itself. Hamman et al. (2010) document a complementary pattern: dictators are more generous when they allocate directly than when they select an intermediary from a set of candidates. Qualitative responsibility measures align with the punishment patterns—dictators feel less responsible when they delegate, and recipients place less blame on principals than on intermediaries.

Coffman (2011) replicates the reduction in punishment under delegation but reinterprets the mechanism. He finds that punishment decreases not because of any inherent shifting of blame to the intermediary, but because the dictator no longer “directly interacts” with the affected party. This holds even when the intermediary is powerless. Oexl and Grossman (2013) extend the design to include various restrictions on the intermediary's choice set and confirm that punishment can be successfully avoided through delegation even to a powerless intermediary, in a setting where punishment is costly, the intermediary's incentives are aligned with the dictator's, and the intermediary can also be punished. Together, the two studies suggest that the lack of direct interaction between dictator and recipient is a key channel through which delegation reduces punishment.

Hill (2015) provides a vignette-based corroboration in a legislative-delegation context. Subjects directly evaluate blame in unincentivised hypothetical scenarios; the resulting attributions track the patterns observed in incentivised experiments, suggesting that punishment captures more qualitative blame attributions reliably. Steffel et al. (2016) push the literature in a complementary direction, showing that people are more willing to delegate decisions made on behalf of others than decisions made for themselves, especially when those decisions might have negative outcomes; the asymmetry is driven both by anticipated responsibility and by anticipated blame, and—a point we return to repeatedly—*people care more about avoiding blame for bad outcomes than about claiming credit for good ones*. Erat (2013) provides a related demonstration that people are willing to pay to delegate the act of lying, consistent with delegation reducing the

psychological cost of behavior that one expects to be judged negatively. Beyond delegation per se, the broader market-design literature documents that intermediation and group decision-making can erode moral concern for third parties: [Falk and Szech \(2013\)](#) show that bilateral and multilateral market interactions substantially increase subjects’ willingness to harm a third party (a mouse) for monetary gain, a canonical demonstration of diffusion of responsibility.

These studies share a common conclusion: in settings with self-interested choice between a fair and an unfair allocation, delegation reduces punishment of the principal, and the prospect of punishment in turn motivates delegation. Their design space, however, has two important features that constrain external validity to algorithmic settings. First, the intermediary is human, and the principal can typically be punished for choosing a particular kind of intermediary. Second, the underlying choice is between a known fair and a known unfair allocation, with no inherent uncertainty in the outcome. Whether responsibility avoidance is just as effective when the intermediary is an algorithm and the underlying decision involves genuine uncertainty is the question we take up. We turn next to what is known about delegation when the intermediary is algorithmic.

Delegation to algorithms

A large literature studies when people use algorithmic decision aids. [Dietvorst et al. \(2015\)](#) introduced the term “algorithm aversion” for the finding that people prefer their own (or other humans’) judgement over algorithmic predictions, even after observing that algorithms outperform human decision-makers. Subsequent work has documented many moderators of this preference. Algorithm aversion is stronger when the algorithm cannot be observed to learn from its mistakes ([Berger et al., 2021](#)), when the algorithm’s reasoning is opaque ([Litterscheidt and Streich, 2020](#); [Sharan and Romano, 2020](#)), when the algorithm is presented as more complex ([Stein et al., 2020](#)), and—most importantly for our purposes—when the underlying decision has moral content or affects third parties ([Gogoll and Uhl, 2018](#); [Bigman and Gray, 2018](#); [Jauernig et al., 2022](#)). Algorithm aversion is also stronger when the task is perceived as more uncertain ([Dietvorst and Bharti, 2020](#)), when delegation removes any ability to override the algorithm’s choice ([Dietvorst et al., 2018](#)), and when the algorithm is used in tasks perceived as subjective ([Castelo et al., 2019](#); [Bogert et al., 2021](#)). [Logg et al. \(2019\)](#), by contrast, document conditions under which subjects show *appreciation* of algorithmic advice over human advice; the picture that emerges is therefore one of substantial heterogeneity in algorithm preferences across tasks, contexts, and elicitation methods. Recent additions include [Bockstedt and Buckman \(2025\)](#), who show that loss-framing of decision tasks eliminates the preference for human assistance over AI assistance, and [Ivanova-Stenzel and Tolksdorf \(2024\)](#), who develop a menu-based design to elicit willingness to cede control to algorithms versus humans. [Burton et al. \(2020\)](#), [Mahmud et al. \(2022\)](#), and [Chugunova and Sele \(2022\)](#) provide systematic reviews of this literature.

Within this body of work, a smaller subset focuses on settings in which the decision being delegated has consequences for someone other than the decision-maker, sometimes called the

moral domain (Gogoll and Uhl, 2018). Gogoll and Uhl (2018) find that subjects in a laboratory experiment preferred to delegate a simple calculation task to another participant rather than to an algorithm even when the task was a straightforward arithmetic exercise where algorithmic advantage should have been salient; subjects had observed the algorithm fail in 80% of cases, however, which complicates inference about pure preferences. Goldbach et al. (2019) report similar aversion when the decision affects both the decision-maker and a third party, and Niszczoła and Kaszás (2020) document the same in a financial decision-making context. Newman et al. (2020) extend the picture to the human-resources domain, showing that algorithmic HR decisions are perceived as less fair than identical human decisions because of “algorithmic reductionism”—the perception that algorithms quantify and decontextualise the qualitative features of performance. The consensus across this subset is that aversion to algorithms is amplified when third-party welfare is at stake. The mechanism, however, is unsettled: *lack of trust in the algorithm*, *loss of perceived control*, and *anticipated reputational or moral costs* have all been proposed, and existing designs typically cannot separate them. We argue in Section 5 that one specific mechanism—*anticipation of judgement when punishment is possible*—deserves a sharper look, since our results show that turning the punishment possibility on or off substantially shifts adoption.

Responsibility perceptions of decisions involving algorithms

A small but growing literature examines how decisions involving algorithms are evaluated by third parties. Most early work is hypothetical and unincentivised: subjects read scenarios involving human-algorithm pairs and rate responsibility on Likert scales. Naquin and Kurtzberg (2004) find that observers attribute less responsibility to a company when an accident involves technology. Pezzo and Pezzo (2006) find similarly that observers attribute less fault to a doctor who follows a (bad) recommendation from an automated decision aid. Liu (2021) replicate this effect in a contemporary setting. Shank et al. (2019) consider moral violations produced by human-algorithm pairs and find that humans who monitor algorithms are blamed less than humans acting alone or humans receiving algorithm recommendations, suggesting that responsibility attributions are sensitive to the precise division of authority between human and machine. Bigman and Gray (2018) document a robust “algorithmic outrage asymmetry”: moral violations by algorithms produce less outrage than identical violations by humans. Tobias et al. (2021) extend the discussion to liability law and find experimentally that physicians who follow standard-care AI recommendations face reduced legal exposure; they argue that this could constitute an incentive to follow such recommendations specifically because doing so shields physicians from blame. The literature thus far suggests that responsibility attributions to humans involved with algorithms differ systematically from attributions to humans acting alone, but the direction depends on the exact configuration.

A smaller incentivised experimental literature has begun to test these ideas in laboratory settings with monetary stakes. Kirchkamp and Strobel (2019) compare a modified dictator game

in which the dictator either acts alone or shares responsibility with a computer; they find no significant differences in self-reported responsibility between human-computer and human-human teams, although a (statistically insignificant) tendency towards more selfish allocations under shared responsibility with humans. [Maasland and Weißmüller \(2022\)](#) find in a computer-aided HR setting that algorithm aversion *dominates* blame avoidance: the prospect of using an algorithm to delegate an unpleasant HR decision (a dismissal) does not overcome the underlying aversion to algorithms in this domain. The most closely related study is [Feier et al. \(2022\)](#), who study delegation to an algorithm in an incentivised laboratory experiment. In their design, subjects first complete ten logic tasks, are randomly assigned to either a “human” or “machine” treatment, and then either rely on their own performance or delegate to (respectively) another human or an algorithm to determine a third party’s payoff. Third parties may then reward or punish. The headline finding is that decision-makers are held *less* responsible for decisions delegated to an algorithm than for decisions delegated to another human, conditional on outcome. Two recent additions push this literature further. [Chevrier and Teixeira \(2024\)](#) extend the blame chain to include the programr behind the algorithm: in their design, AI programrs and the delegator can both escape responsibility for inegalitarian AI-mediated allocations, creating a “moral wiggle room” that incentivises programrs to push the AI toward more inegalitarian choices. [Köbis et al. \(2025\)](#), in a high-profile recent study in *Nature*, document that humans request unethical behavior from machine agents more readily than from human agents, in part because the machines (large language models including GPT-4, Claude 3.5 Sonnet, and Llama 3.3) comply with such instructions 58–98% of the time, compared to 25–40% for human agents. Their setting (active dishonesty under instruction) differs from ours (delegation under outcome uncertainty), but the AI-as-blame-shield motif is shared.

The [Feier et al. \(2022\)](#) design, which is the immediate precedent for ours, has three features that complicate the inference about pure responsibility shifting. First, the algorithm in their experiment is calibrated at the session level rather than at the individual level: each subject is shown the distribution of algorithmic performance across the relevant pool, and her delegation decision then reflects beliefs about her own ability *relative* to that distribution. The delegation choice therefore carries strong signal value about Player A’s self-perception, which contaminates downstream punishment as a measure of responsibility shifting per se. [Feier et al. \(2022\)](#) indeed report that the delegation decision is well explained by perceived own performance relative to the algorithm. Second, the same subject acts as both delegator and evaluator, introducing additional motives—self-justification, hypocrisy aversion, projection—into the punishment decision. Third, their design does not include a treatment in which punishment is ruled out by construction, so the question of whether the punishment possibility itself motivates delegation cannot be addressed in their data.

Theoretical microfoundations: identity, moral self-image, and process ownership

The mechanism we adopt in interpreting our results—a desire to be the proximate cause of a good outcome that is amplified by the moral salience of accountability—draws on two adjacent theoretical traditions. The *identity / moral self-image* literature in economics, anchored by [Bénabou and Tirole \(2011\)](#), models people as caring about “who they are” and inferring their own values from past choices. Identity functions as an asset that responds non-monotonically to acts and threats, producing asymmetric reactions to good versus bad outcomes that map naturally onto the credit-blame asymmetry documented by [Steffel et al. \(2016\)](#). The *psychological-ownership* literature, in turn, has documented that customising or actively choosing an output transfers a sense of ownership to it: people feel more responsible for outcomes they directly bring about than for those they delegate. The intersection of these two literatures with our setting is largely unexplored: no formal model directly links the identity framework to algorithm-mediated decisions. Our M3 mechanism (developed in Section 4) is in this sense a hypothesis derived from the identity framework rather than a confirmed account, and we treat it as such throughout.

Where this paper fits

Our experiment is the first to our knowledge to (i) calibrate the algorithm to mimic each individual subject’s performance, eliminating the relative-performance confound; (ii) separate the role of delegator and evaluator across distinct subject pools; and (iii) include a between-subject manipulation that turns the punishment possibility on or off, providing a clean test of whether anticipated punishment is a motive in the delegation decision itself. The first two features sharpen the test of the responsibility-shifting hypothesis (H2) by removing alternative interpretations of differential punishment. The third moves beyond the existing literature, which has documented punishment differentials *conditional on* the punishment possibility but has not directly identified whether the threat of punishment causally moves the delegation decision.

Reconciling our findings with [Feier et al. \(2022\)](#). Two of our results sit in apparent tension with the closest precedent. First, [Feier et al. \(2022\)](#) document that decision-makers are punished *less* for algorithmically delegated decisions; we find no significant punishment differential. Second, the natural prediction from [Feier et al. \(2022\)](#)’s own findings is that the prospect of punishment should make algorithmic delegation *more* attractive; we find the reverse. We do not view these contrasts as contradictions, but as evidence that the design features which differ between the two studies are doing real work. The differential punishment in [Feier et al. \(2022\)](#) is most plausibly read as third parties extracting a signal from the delegation decision about the principal’s self-perceived ability relative to the algorithm—a signal that is

maximally informative when (as in their design) the algorithm is calibrated at the session level and when the principal’s relative ability is uncertain. Our individual-level calibration, made common knowledge, closes off this signal-extraction channel by construction: a Player B in our setting has no basis to read delegation as overconfidence, laziness, or ability avoidance. The disappearance of the punishment differential in our data is therefore consistent with, rather than refutes, the signal-extraction reading of [Feier et al. \(2022\)](#)’s result. By the same logic, the responsibility-shifting motive that [Feier et al. \(2022\)](#)’s setup leaves on the table is muted in ours, isolating a different and milder margin: the moral salience of accountability for the principal herself, independent of any third-party inference. The reversal we observe in the delegation decision speaks to that margin—and we are explicit that the question of which design feature is causally responsible for the sign reversal cannot be resolved without an experiment that varies relative-performance calibration directly, which we do not do here.

We see scope, then, for the two studies to be parts of a single picture: [Feier et al. \(2022\)](#) identifies the signal-extraction channel through which delegation can shift third-party blame; we identify a process-ownership channel through which the prospect of accountability can pull the principal toward direct engagement with the decision. The two channels are not mutually exclusive. [Chevrier and Teixeira \(2024\)](#), looking at a third margin (the programr behind the algorithm), and [Köbis et al. \(2025\)](#), looking at a fourth (the compliance asymmetry between human and machine agents on instructed unethical tasks), populate further nodes in this expanding map. A unified design that varies the relative-performance calibration, the punishment possibility, and the visibility of the algorithm’s compliance properties would be the natural next step. We discuss the implications for future work in [Section 5](#).

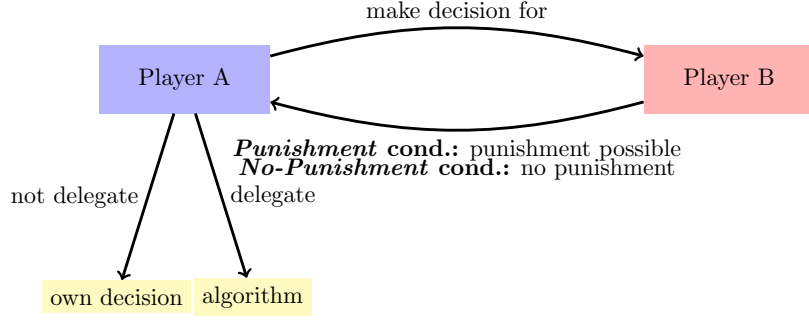
3 Experimental Design

We conducted an online experiment, preregistered at AsPredicted #133980 (aspredicted.org/hs9az8.pdf), on the British platform Prolific between February and June 2023, recruiting 322 participants from the United Kingdom. The design combines a between-subject manipulation of whether costly punishment is possible with a within-subject elicitation of conditional punishment behavior.

Within each condition, participants assumed one of two roles. Player A decides whether to make a payoff-relevant prediction for Player B directly, or to delegate the prediction to an algorithm calibrated to perform as well as she does. Player B is the matched recipient whose payoff is fully determined by the outcome of Player A’s prediction; in the *Punishment* condition Player B may subsequently punish Player A, conditional on both the delegation decision and the realized outcome, while in the *No-Punishment* condition no punishment is possible. [Figure 1](#) summarizes the basic structure.

We targeted a balanced design of 80 participants per role per condition, for a planned sample

Figure 1: Conditions overview



of 320 subjects.¹ The experiment was programmed in oTree (Chen et al., 2016) and run via Prolific. Eligibility followed Prolific’s standard quality screens—minimum 95% prior approval rate, no warnings on file, and a fluent-English filter—together with a desktop-only restriction (no mobile or tablet) to keep the screen layout consistent across subjects.

Procedure

Player A

After providing informed consent, Player A completed ten incentivized rounds of a visual prediction task. In each round, she predicted a person’s weight from a facial image and the person’s height. Predictions could be entered via slider or typed directly in kilograms, pounds, or stones-and-pounds; no feedback was provided between rounds.² At the end of the experiment, one of the ten rounds was randomly selected for payment: Player A earned £4 if her prediction in that round was within 10 lbs of the true weight and £2 otherwise. These initial rounds served three purposes simultaneously: they established a stake from which any later punishment could be deducted; they familiarized Player A with the task; and they generated the performance data needed to calibrate the algorithm.

The choice of a visual prediction task is deliberate. Algorithm preferences depend sensitively on the algorithm’s perceived relative competence (Dietvorst et al., 2015; Logg et al., 2019);

FB CHECK: Verify that Dietvorst (2015) and Logg (2019) support the “perceived relative competence” framing. Dietvorst (2015) is largely about confidence loss after observing errors rather than about ex-ante perceived competence; Logg (2019) is about algorithm appreciation as a function of perceived ability.

as described below, our design neutralizes this concern by calibrating the algorithm to match each subject’s individual performance. The visual nature of the task makes algorithmic error plausible and lends credibility to the equal-performance claim, in contrast to purely mathematical tasks where subjects may be skeptical that an algorithm could ever fail (Gogoll and Uhl,

¹Appendix C reports the full ex-ante minimum-detectable-effect calculations underlying this cell size.

²All individuals depicted in the images had explicitly consented to the use of their photographs for research purposes.

2018).

FB CHECK: Verify that Gogoll & Uhl (2018) actually support the math-task scepticism contrast. The paper’s main claim is about moral-domain aversion; the math-task framing may not be in the paper.

Predicting body weight from a facial image has been used in earlier experiments (Logg et al., 2019) and arguably levels the playing field between human and algorithm: humans have a lifetime of practice at this mapping, while algorithms cannot easily exploit large training corpora here. Finally, we chose a prediction task involving genuine uncertainty rather than a setting with an obvious self-serving motive (as in classic dictator-game variations). This reflects the nature of many real-world applications of algorithms—financial forecasting, medical diagnosis, fraud detection—in which outcomes are inherently uncertain and performance is evaluated ex post.

After the ten rounds, Player A was randomly paired with another participant, Player B. Player A was informed that her final prediction would not affect her own payoff but would almost entirely determine the payoff of Player B, who had not completed the prediction task himself: if the prediction fell within 10 lbs of the true weight, Player B would receive a high payoff (£5); otherwise a low payoff (£1). Player A then learned that she could either make the prediction herself or delegate it to an algorithm calibrated to mimic her own performance from the first ten rounds, so that on expectation the algorithm would produce the high payoff for Player B at the same rate as Player A would.

Concretely, the algorithm draws a Bernoulli outcome with success probability equal to Player A’s empirical accuracy rate over her first ten rounds. A subject who scored 3/10 in the initial rounds thus delegates to an algorithm that produces the high payoff for Player B with 30% probability. This individual-level Bernoulli calibration is the central design innovation relative to the closest precedent (Feier et al., 2022), and it does substantive work in the identification argument. First, it ensures the algorithm’s success probability equals Player A’s empirical accuracy over the first ten rounds for every subject, making the equal-performance claim true in the data on which the calibration is based. Second, it strips any signal value that the delegation decision might otherwise carry about Player A’s beliefs concerning her own ability relative to the algorithm: a Player B observing the delegation decision has no basis on which to read it as overconfident, lazy, or selfish. This closes off the most plausible third-party signal-extraction channel through which delegation might otherwise insulate the principal from punishment.

The contrast with session-level calibration is consequential. Had the algorithm instead been calibrated to reflect aggregate performance across a pool of participants—as in Feier et al. (2022)—the delegation choice would be confounded by each subject’s beliefs about her own ability relative to the pool. Indeed, Feier et al. (2022) report that the delegation decision in their setting is largely driven by perceived own performance relative to the algorithm. Pairing the algorithm to each subject individually, and making this common knowledge between Player A

and Player B, strips this confound out by construction.

In the *Punishment* condition, Player A was further informed that Player B could reduce her payoff by up to £2, with the deduction depending on both her delegation decision and the realized outcome. In the *No-Punishment* condition, Player B had no punishment option. The two delegation options (own decision / algorithm) were presented in random order across subjects. We attached no monetary cost to delegation. This brings us closer to Player A’s intrinsic preference over delegation in the *No-Punishment* condition, since any positive cost would confound the responsibility-shifting motive with a price effect; costless delegation is standard in the responsibility-shifting literature for this reason (Bartling and Fischbacher, 2012; Coffman, 2011; Feier et al., 2022).

We then asked Player A to report her beliefs about Player B’s punishment in each of the four scenarios resulting from the combination of delegation decision (yes/no) and prediction outcome (high/low payoff for Player B). Beliefs were elicited on the same scale as actual punishment (£0–£2 in £0.10 increments) and the four scenarios were presented in block-wise random order. We chose not to incentivize belief elicitation for three reasons. First, only two of the four scenarios remain payoff-relevant after the delegation decision, making the others purely hypothetical. Second, an incentivized mechanism could induce hedging behavior. Third, an incentive-compatible scoring rule across four scenarios would have substantially increased task complexity. Because we are nonetheless interested in comparing punishment beliefs to actual punishment, we included an attention check on the belief-elicitation screen.³ In the *No-Punishment* condition, no punishment was possible; we elicited beliefs about hypothetical punishment to maintain consistency across treatments.

Player A was then asked to assess her own performance in the first ten rounds by assigning a probability mass (summing to 100%) to each possible score from 0 to 10. We elicited this belief after the delegation decision, and after Player A had learned of the equal-performance calibration, for three reasons: the post-update belief is the analytically relevant input for any subsequent behavior; pre-decision elicitation would risk introducing measurement-induced anchoring on the delegation decision itself; and the elicitation is used as a moderator rather than as a determinant of delegation in the primary analysis. The elicitation was incentivized: Player A earned an additional £0.30 with the probability she had assigned to her actual score.⁴ Player A was also asked to estimate the population distribution of scores by assigning a share to each possible score from 0 to 10, which lets us construct a measure of perceived relative ability that the prior literature’s pooled-calibration designs cannot identify. One score was randomly drawn at the end of the experiment and Player A earned £0.30 if her estimate for that score was within five percentage points of the realized population share.

Finally, we elicited risk preferences using an incentivized multiple price list in the style of Holt and Laury (2002) and administered a short questionnaire on socio-demographic characteristics

³Subjects who failed the attention check are excluded from belief-related analyses.

⁴The mechanism is incentive-compatible only for risk-neutral subjects; in practice most subjects spread probability mass across several scores.

and technology affinity, before presenting Player A with her preliminary payoff. Figure 8 in Appendix F summarizes the experimental timeline for Player A.

Player B

After providing informed consent, Player B was told that he had been randomly matched with a Player A and was introduced to the task Player A had completed. He learned that his own payoff would be determined by the outcome of Player A’s final prediction—£5 if it fell within 10 lbs of the true weight and £1 otherwise—while Player A’s payoff depended solely on her performance in the initial ten rounds. Crucially, Player B was also told that Player A could delegate the final prediction to an algorithm calibrated to perform equally well as Player A in the earlier rounds, and that Player A herself was aware of this equal-performance calibration. The equal-performance claim is therefore common knowledge between the two players, a feature on which our identification argument rests.

In the *Punishment* condition, Player B could choose to reduce Player A’s payoff by up to £2 in £0.10 increments. We focus on punishment rather than reward because the responsibility-attribution literature has predominantly operationalized blame through sanction-imposing behavior (Bartling and Fischbacher, 2012; Coffman, 2011; Steffel et al., 2016), and because the applied policy questions about algorithmic accountability that motivate our paper are likewise framed in sanction-language. Imposing strictly positive punishment in any cell required Player B to forgo £0.10 of his own earnings for that cell, ensuring that punishment was costly to the punisher.⁵ We elicited punishment decisions for all four combinations of delegation (yes/no) and outcome (high/low payoff for Player B) via the strategy method, implementing only the decision matching the realized combination; the strategy method gives us the within-subject power to detect cell-level differences with the achieved sample size.⁶ Because the punishment decision is the central outcome variable, we included an attention check on the punishment-elicitation screen.⁷ In the *No-Punishment* condition, where actual punishment was not possible, we elicited hypothetical punishment decisions on the same scale to keep the experimental flow comparable across conditions. Player B’s perceptions of task difficulty, his risk preferences, and a short questionnaire followed, before he was presented with his final payoff. Figure 9 in Appendix F summarizes the experimental timeline for Player B.

Once Player B’s decisions had been recorded, Player A’s final payoff was calculated and all participants were paid on the day of participation. Figure 7 in Appendix E reports the experimental flow of participants by condition and role, including recruitment, attention-check exclusion, and the analysis sample.

⁵The cost is fixed per scenario rather than proportional to the punishment level, so that strictly positive punishment is always costly relative to no punishment but the cost does not increase with the severity of the sanction.

⁶The strategy method may yield different absolute punishment levels than direct elicitation; we discuss the implications and a possible direct-method companion as future work in Section 5.

⁷The check asked the subject to confirm a numerical aspect of the payoff structure that had been displayed on the previous screen. Subjects failing this check are excluded from punishment-related analyses.

Hypotheses and identification

Our design tests two preregistered hypotheses, both grounded in the literature on responsibility avoidance through delegation (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013; Hamman et al., 2010; Hill, 2015; Steffel et al., 2016) and on responsibility attributions to algorithmic decision-makers (Feier et al., 2022; Kirchkamp and Strobel, 2019; Gogoll and Uhl, 2018).

H1 (Delegation): The introduction of potential punishment increases the likelihood that Player A delegates the decision to the algorithm.

H1 is grounded in empirical evidence that responsibility avoidance shapes delegation decisions involving human intermediaries. Bartling and Fischbacher (2012) also show that subjects are more likely to delegate morally consequential decisions even to a die roll—an early form of delegation to a non-intentional agent—suggesting that responsibility avoidance persists when the agent lacks intentionality. Whether the same logic extends to *algorithmic* delegates is less settled. Feier et al. (2022) report that delegation rates are similar for human and algorithmic intermediaries, suggesting that the *kind* of intermediary may matter less than the mere availability of one—but their design does not directly compare delegation rates across punishment regimes, leaving open the question H1 addresses.

FB CHECK: Felix asked whether Feier et al. (2022) report whether delegation propensity is HIGHER under punishment than without for both AI and human delegates. Their design does not appear to directly run this between-treatment comparison, but please verify in the paper.

H1 is identified by the between-subject comparison of Player A’s delegation rate across the *Punishment* and *No-Punishment* conditions, which exploits the punishment manipulation as the only between-subject difference and holds fixed every other feature of the decision environment.

H2 (Punishment): Conditional on a low-payoff outcome for Player B, Player A is punished less when she delegated the decision to the algorithm than when she made it herself.

H2 mirrors H1 on the punishment side. Bartling and Fischbacher (2012) find that delegation to a randomization device still allows decision-makers to avoid some punishment, although less so than delegation to another human. Feier et al. (2022) find that decision-makers are rewarded less when a negative outcome is directly attributable to their own action, compared to when it results from an algorithm to which they delegated. Their study differs from ours in three respects, however: a different task; an emphasis on reward differentials rather than punishment; and—most consequentially—a design that does not control for relative performance at the individual level. As a result, in their setting the delegation decision carries a strong signal about Player A’s beliefs concerning her own ability relative to the algorithm; a failure to

delegate may then be punished as overconfidence rather than as direct responsibility-taking, contaminating the inference about responsibility-shifting per se. H2 is identified by the within-subject comparison across the four (delegation, outcome) cells of Player B’s strategy-method punishment, which isolates the conditional-on-outcome effect of delegation on punishment, holding fixed both the realized consequences for Player B and all between-subject heterogeneity in Player B’s punishment proclivity.

Two further design features deserve note. First, although the prediction task affects Player B’s payoff, the incentives of the two players are nominally aligned: Player A has no direct financial interest in Player B receiving the low payoff, and a successful prediction strictly weakens her own expected punishment.⁸ This alignment mirrors canonical real-world settings for algorithmic delegation—medical diagnosis, financial forecasting, judicial decision-support—in which the decision-maker has no direct material interest in the affected party’s outcome but bears reputational or moral consequences for poor outcomes; by replicating this alignment, our design speaks directly to the policy-relevant cases. A standard intention-based punishment motive should therefore have less bite here than in dictator-game settings where the decision-maker can choose to be selfish. Second, although the underlying decision (a prediction under uncertainty) is not itself moral, the setup nonetheless qualifies as part of the *moral domain* as defined in the algorithm-aversion literature (Gogoll and Uhl, 2018): Player A’s choice imposes monetary externalities on a matched recipient, and a poor decision causes real, albeit modest, monetary harm. The implications of these two design choices for the interpretation of our results are taken up in Section 4.

4 Results

Data collection took place between February and June 2023 through the British online platform Prolific, with all participants recruited from the United Kingdom.⁹ A total of 322 subjects were recruited (161 assigned to Player A, 161 to Player B).¹⁰ On average, Player A spent 12 minutes completing the experiment and earned an average bonus of £2.49 on top of the Prolific show-up fee, while Player B spent approximately 8 minutes and earned an average bonus of £2.30. Treatment randomization was effective, as demonstrated by the balance across observable characteristics (Appendix A.1, Table 2). A flow diagram showing recruitment, attrition, attention-check exclusion, and the analysis sample, broken down by condition and role, appears in Appendix E; in particular, attrition rates do not differ meaningfully across the *Punishment* and *No-Punishment* conditions.

We preregistered exclusion of subjects who failed the attention check on the screen at which

⁸In the *Punishment* condition, a better outcome for Player B reduces expected punishment for Player A; in the *No-Punishment* condition there is no punishment to reduce. Either way, Player A’s incentives are weakly aligned with Player B’s.

⁹EXCLUSION CRITERIA: leave a comment for Felix to insert later

¹⁰We preregistered the collection of data from 320 participants (80 subjects per role per treatment). Due to attrition and subsequent re-matching we recruited an extra pair in the *Punishment* condition.

their main outcome variable was elicited. For Player A, this is the belief-elicitation screen; for Player B, the punishment-elicitation screen. Of 161 Player Bs, 135 pass the attention check (83.9%); the corresponding pass rate for Player A is 131/161 (81.4%). All main results are robust to including subjects who fail the attention check; we report these robustness checks in Appendix A.2.

FB: maybe move to first set of results. Also the appendix does not show anything and the default should be including everyone.

Result 1: Punishment reduces delegation

Headline. In the *Punishment* condition, where Player B can punish, 42.5% of Player As delegated the decision to the algorithm. In the *No-Punishment* condition, where punishment is impossible, 59.3% delegated. Figure 2 illustrates the difference. The introduction of potential punishment thus significantly *reduces* the likelihood of delegation to the algorithm (Pearson Chi-squared, $p = 0.033$; Fisher’s exact, $p = 0.041$).¹¹¹²

Result 1. *The possibility of punishment significantly reduces the likelihood that Player A delegates the decision to the algorithm.*

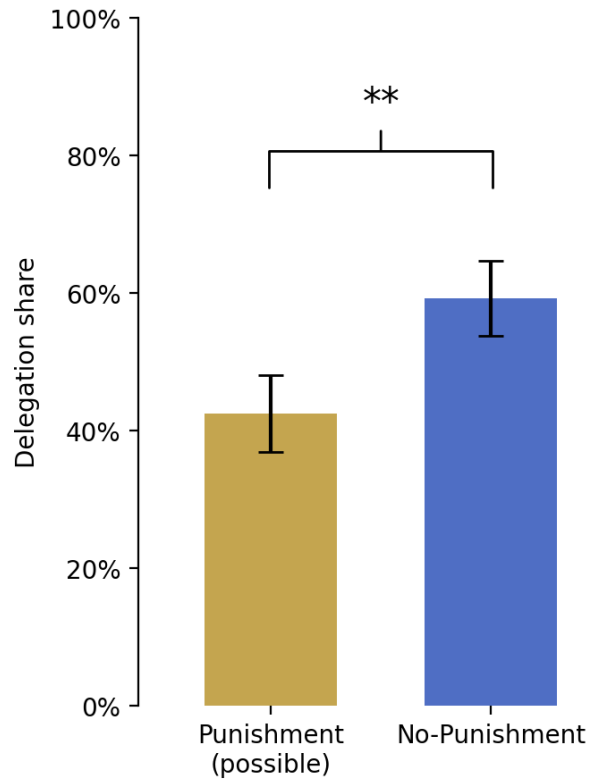
This result not only fails to support *Hypothesis 1*; it points in the opposite direction. The responsibility-avoidance literature, on which H1 rests, would predict the introduction of punishment to make delegation *more* attractive as a tool for shifting responsibility. We instead observe a sizeable reduction in delegation when punishment is on the table. Before turning to candidate mechanisms for this reversal in Section 4, we first establish that the effect is robust and document its empirical structure.

With controls. Table 1 reports estimates from a Logit model of the delegation decision. Column (1) reports the unconditional treatment effect; Column (2) adds Player A’s task performance and a vector of socio-demographic and technology-affinity controls. The treatment effect is robust: the coefficient on the *No-Punishment* indicator is 0.68 ($p = 0.034$, two-sided) in the unconditional specification and 0.76 ($p = 0.023$, two-sided) with controls. The full set of estimated coefficients is reported in Appendix Table 3.

¹¹The preregistration is non-directional on H1; the realized direction is opposite to that anticipated by the responsibility-avoidance literature on which the hypothesis was grounded. We report two-sided tests throughout the body of the paper. The full set of preregistered tests, the deviations from each, and the rationale appears in Appendix D.

¹²A within-subject corroborator points the same direction. Player As in the *No-Punishment* condition were asked, after their actual delegation decision, the hypothetical question of whether they would have delegated had punishment been possible. Among the 81 respondents, 12 would have stopped delegating under hypothetical punishment while only 5 would have started, yielding a hypothetical delegation rate of 50.6% (within-subject McNemar’s test $p = 0.090$). Hypothetical responses elicited after the actual decision are subject to motivated reasoning; the within-subject pattern complements rather than substitutes for the between-subject test.

Figure 2: Delegation shares by condition



Note: Bars show the share of Player As who delegated the final prediction to the algorithm in each between-subject condition. The two-sided difference is significant at the 5% level under both a Pearson Chi-squared test ($p = 0.033$) and a Fisher's exact test ($p = 0.041$). The preregistration does not specify a test direction; the realized effect reverses the sign anticipated by the responsibility-avoidance literature on which the hypothesis was grounded.

Table 1: Determinants of the delegation decision

	Dependent variable: <i>Delegation</i>		
	<i>Full sample</i>		<i>Punishment only</i>
	(1)	(2)	(3)
No-Punishment indicator	0.677 (0.320)	0.757 (0.334)	
Punishment beliefs: bad outcome (no del. – del.)			0.937 (0.751)
Punishment beliefs: good outcome (no del. – del.)			1.579 (0.741)
Task performance		–0.192 (0.128)	–0.609 (0.236)
Controls	no	yes	yes
N	161	159	60
Pseudo R^2	0.020	0.049	0.246

Note: Coefficients from a Logit regression of the delegation decision on the condition indicator, task performance, and socio-demographic controls. Robust (HC1) standard errors in parentheses. Two-sided p -values for the headline coefficients: *No-Punishment indicator* Col (1) $p = 0.034$; Col (2) $p = 0.023$; *Task performance* Col (3) $p = 0.010$; *Punishment beliefs: good outcome* Col (3) $p = 0.033$. Controls in Columns (2) and (3) include age, gender, self-reported socio-economic status, university education, technology affinity, and leadership experience; full results in Appendix Table 3. Two subjects are dropped in Column (2) due to missing socio-demographic data. Column (3) restricts to the *Punishment* condition and includes within-subject differences in expected punishment between non-delegated and delegated decisions for each outcome; subjects who failed the belief-screen attention check are excluded, as preregistered. Following AEA editorial policy, significance stars are not reported in the table; exact p -values are stated in this note.

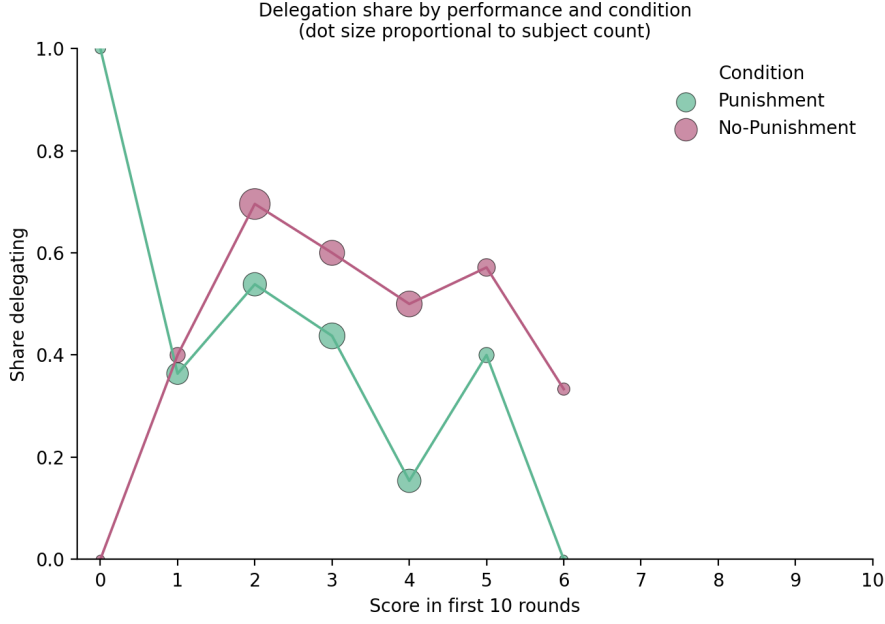
Robustness. The treatment effect survives several further checks. First, controlling for Player A’s overconfidence (defined as the weighted-average belief about own performance minus actual performance) leaves the *No-Punishment* coefficient essentially unchanged (0.77, $p = 0.022$); overconfidence itself is not significantly associated with delegation ($p = 0.272$). Second, order effects are localised to the *Punishment* condition: presenting the algorithm option first roughly doubles delegation (52.5% vs 32.5% when the own-decision option is first; χ^2 $p = 0.070$), while the *No-Punishment* condition shows no order effect ($p = 0.674$); the treatment $\times$ order interaction is not significant ($p = 0.115$), so the H1 finding is not driven by interaction with order. Third, relaxing the preregistered attention-check exclusion does not change the result. The full robustness specification is reported in Appendix Table 4.

Heterogeneity by performance. Within the *Punishment* condition, Player As who performed better on the first ten rounds were less likely to delegate. The Pearson correlation between the score in the first ten rounds and the probability of delegation is -0.19 (two-sided $p = 0.092$); the corresponding regression coefficient in Column (3) of Table 1 is -0.61 (two-sided $p = 0.010$). The strengthening of the relationship in the regression reflects, in part, the attention-check restriction (Column (3) is identified off $n = 60$ Punishment-attention-pass subjects, while the raw correlation uses all 80 Punishment subjects). In the *No-Punishment* condition, no comparable relationship between performance and delegation is observed. Figure 3 visualises this asymmetry. We flag this paragraph as the empirical hint that motivates the process-ownership mechanism developed in Section 4: the condition effect is concentrated among Player As who, conditional on punishment being on the table, prefer to take a personal stake in producing a good outcome for Player B.

Result 2: We cannot reject equal punishment of delegated and self-made decisions

Headline. We now turn to Player B’s punishment behavior in the *Punishment* condition. Punishment in all four (delegation, outcome) cells of the strategy method is significantly greater than zero, and Player Bs punish more for low-payoff outcomes than for high-payoff outcomes—an indication that punishment carries informational content rather than being noise; 41.8% of Player Bs nonetheless do not punish in any of the four scenarios. Conditional on the outcome, however, Player A is not significantly differently punished depending on the delegation decision. When Player B receives the high payoff, average punishment is £0.311 for delegated decisions and £0.310 for self-made decisions—a difference of one-tenth of a penny. When Player B receives the low payoff, average punishment is £0.494 when Player A delegated versus £0.571 when she made the decision herself; this difference points in the H2 direction but is far from significant (paired t -test $p = 0.153$; Wilcoxon signed-rank $p = 0.533$). The same picture obtains when measuring punishment frequencies rather than amounts: Player Bs neither punish delegated decisions more harshly, nor do they punish them more frequently (Appendix Figure 10).

Figure 3: Delegation share by performance and condition



Note: Each point reports the share of Player As at the corresponding first-ten-rounds score who delegated the final prediction. Dot size is proportional to the number of subjects at that score. The downward slope in the *Punishment* condition is the Pearson $r = -0.19$ ($p = 0.092$); no comparable slope in *No-Punishment*. Restricted to attention-check passers.

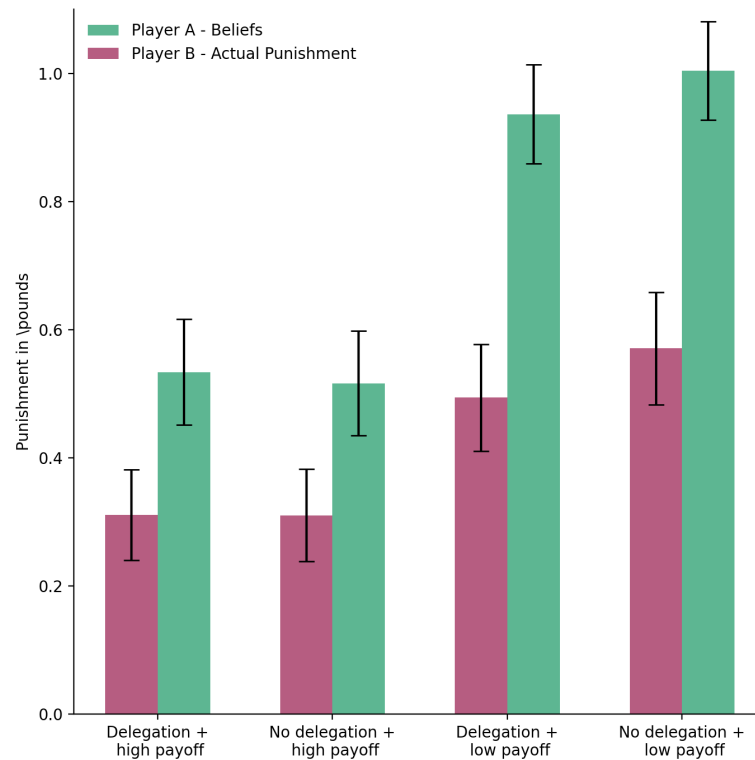
As preregistered, the within-subject test is identified off the 67 Player Bs in the *Punishment* condition who passed the attention check; the result is essentially identical when these subjects are included (mean difference £0.071 vs £0.077; paired t -test $p = 0.131$ vs $p = 0.153$; Appendix Table 6).

Result 2. *Conditional on the outcome for Player B, we cannot reject the hypothesis that Player A is punished equally for delegated and self-made decisions. A TOST equivalence test rules out moderate insulation effects (those exceeding £0.20, or 10% of the maximum punishment) at $\alpha = 0.05$, but we cannot rule out smaller insulation effects.*

Equivalence (TOST). We can refine the negative result with a positive equivalence claim. A two one-sided test (TOST) at the $\pm$ £0.20 bound—against the null that the true insulation effect exceeds 10% of the maximum punishment—rejects in both the bad-outcome cell ($p = 0.012$) and the good-outcome cell ($p < 0.001$). At a tighter $\pm$ £0.10 bound, equivalence cannot be established ($p = 0.332$ for the bad-outcome cell). In our setting we can therefore rule out moderate insulation effects (above £0.20 per cell) with high confidence, but cannot rule out small ones below this bound.

What predicts punishment? Appendix Table 5 reports a regression of chosen punishment on a delegation indicator, an outcome indicator, their interaction, Player B’s beliefs about Player A’s performance, and a vector of socio-demographic controls. The interaction term is small and statistically insignificant ($p = 0.137$), confirming the bivariate result. Punishment

Figure 4: Punishment by Player B and beliefs about punishment by Player A



Note: Average punishment chosen by Player B (solid bars) and average belief about punishment held by Player A (hollow bars) for each of the four scenarios. Bars are reported in pounds and standard errors are reported as error bars. Beliefs are elicited prior to learning the realized outcome and on the same scale as the punishment decision; the belief overlay is analysed in Section 4.

is also independent of Player B’s belief about Player A’s performance: across all four cells, Pearson correlations between cell-specific chosen punishment and Player B’s weighted-average belief about Player A’s score are essentially zero ($|r| \leq 0.03$, all $p > 0.8$). This rules out the alternative explanation that Player Bs who think Player A is more capable systematically punish less (or more); punishment is not driven by performance-conditional inference about Player A.¹³

Interpretation. The fact that we cannot reject equal punishment for delegated and self-made decisions is consistent with our specific design: because the algorithm is calibrated at the individual level and this calibration is common knowledge, the delegation decision in our setting carries no signal about Player A’s belief in her own ability relative to the algorithm. A natural intention-based punishment motive is therefore muted: Player B has no obvious basis on which to read the delegation decision as overconfident, lazy, or selfish. By the same token, however, this design choice limits the test of H2 to a setting in which the most studied path through which delegation might insulate from punishment—signal extraction by the third party—is closed off by construction. The complementary reading is that responsibility shifting through delegation may operate more strongly when relative performance is uncertain (as in [Feier et al. 2022](#)); we discuss in Section 5 why a follow-up study that varies relative-performance calibration directly is the natural way to map this margin.

Mechanism: candidate explanations for the H1 reversal

The direction of Result 1 reverses the standard prediction of the responsibility-avoidance literature, in which the prospect of punishment should make delegation *more* attractive, not less. Because the reversal was not preregistered, the analyses in this subsection are exploratory and we treat the supported mechanisms as candidate hypotheses rather than as established. We consider four candidates—anticipated differential punishment (M1), effort to outperform the algorithm (M2), process ownership (M3), and experimenter demand (M4)—and weigh each against the data, taking care to distinguish what we can rule out from what we can only flag as consistent. Before turning to the four candidates, we lay out the belief evidence on which several of them rely, separating beliefs about Player A’s task performance from beliefs about Player B’s punishment.

¹³Post-experiment Likert measures provide a tentative attitudes–behavior link. Among the 44 Player Bs who passed the attention check and completed the post-experiment questions, those who reported that Player A should feel more responsible for the decision punish self-made good outcomes more relative to delegated good outcomes (Pearson $r = 0.36$, $p = 0.015$ between the responsibility Likert and the within-subject difference $\text{punish_nodeL_good} - \text{punish_del_good}$); the corresponding pattern in the bad-outcome cell goes in the same direction but is not significant ($r = 0.24$, $p = 0.114$). When asked directly who should be accountable for AI-generated decisions, 66% of these subjects identified deploying companies as primarily accountable, 18% developers, 14% end-users, and only one named government or regulatory bodies. The sub-sample is small ($n = 44$); we treat these patterns as suggestive.

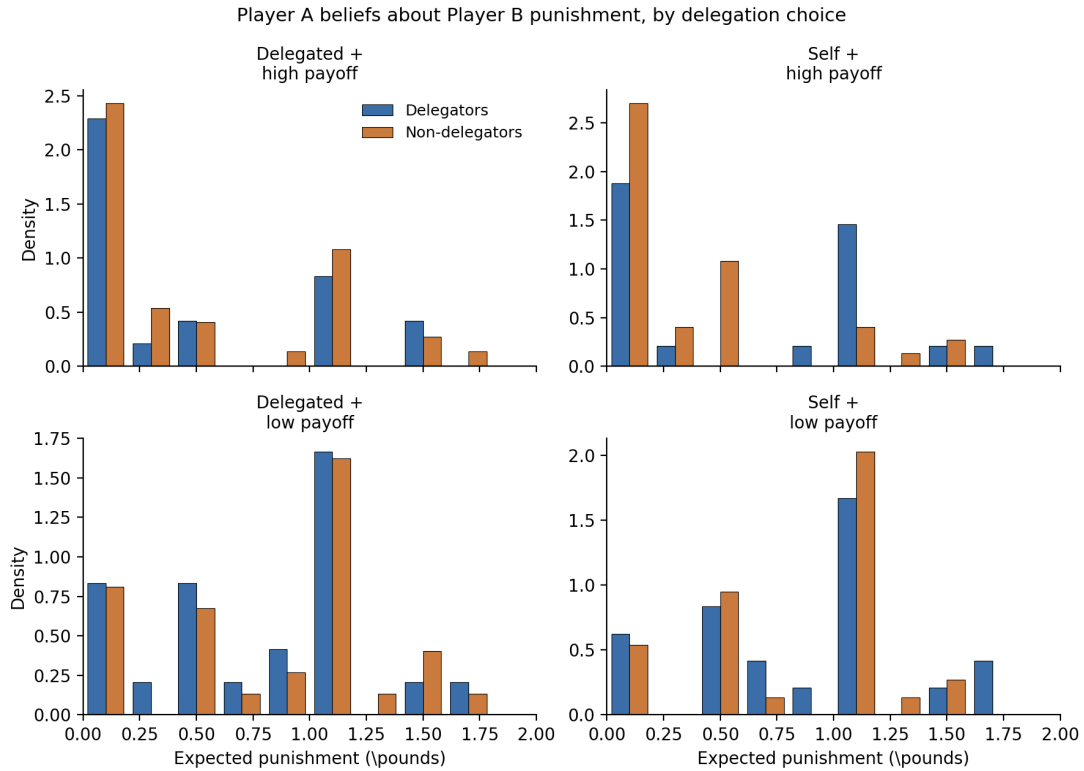
Performance beliefs. Both players overestimate Player A’s task performance. The true mean overall score in the first ten rounds is 2.93/10, while Player A’s weighted-average belief about her own score is 4.76 (overestimate of 1.84 score points) and Player B’s weighted-average belief about Player A’s score is 4.32 (overestimate of 1.40 score points). The overestimation is therefore not a Player-A-specific or Player-B-specific artefact, but a feature of the elicitation. Performance beliefs do not enter the M1 ruling-out argument below; we report them here as a sanity check on the elicitation rather than as evidence on the mechanism.

Punishment beliefs. After making the delegation decision but before learning the outcome, we elicited Player A’s beliefs about Player B’s punishment in each of the four scenarios. Two patterns emerge. First, Player As substantially *overestimate* the level of punishment they will face: average expected punishment exceeds average realized punishment in every scenario (Figure 4, hollow bars). Second, however, Player As correctly anticipate the *relative* pattern of punishment: they expect no significant difference in punishment between delegated and non-delegated decisions, conditional on outcome—a pattern that closely tracks the actual punishment behavior of Player B documented in Result 2. *It is this relative-pattern finding—on punishment beliefs alone—that rules out a simple anticipation mechanism for Result 1.* If Player A had reduced delegation in the *Punishment* because she anticipated being punished more harshly for delegation than for self-made decisions, we would expect to see this asymmetry in the elicited beliefs. We do not.

Within-subject consistency and asymmetric belief patterns. Despite the weakness of the elicitation, punishment beliefs carry within-subject information. Column (3) of Table 1 reports a Logit regression of delegation on Player A’s belief difference (*no delegation* minus *delegation*) for each outcome, restricted to the *Punishment* condition. Coefficients on both belief differences are positive: a Player A who expects to be punished more for non-delegation than for delegation in a given outcome is more likely to delegate. The coefficient on the good-outcome belief difference is significant at conventional levels (1.58, two-sided $p = 0.033$); the coefficient on the bad-outcome belief difference is positive but noisier (0.94, two-sided $p = 0.21$). The within-subject signal is asymmetric across cells (Figure 5): the single significant difference between delegators and non-delegators is in the *self decision + good outcome* cell, where delegators expect punishment of £0.82 while non-delegators expect only £0.32 (t -test $p = 0.006$, Mann-Whitney $p = 0.019$); for the other three cells the gap is small and not significant. Because beliefs were elicited after the delegation decision and the elicitation was unincentivised, we cannot rule out that subjects rationalised their choice by holding consistent post-hoc beliefs; the within-subject correlation is consistent with the standard interpretation that anticipated punishment shapes delegation, but does not identify the causal direction.

(M1) Anticipated punishment for delegation. A Player A who expects to be punished more harshly for a delegated decision than for a self-made decision will rationally delegate less

Figure 5: Player A beliefs about Player B punishment, by delegation choice



Note: Distribution of Player A's expected punishment from Player B in each of the four (delegation, outcome) cells, separately for Player As who actually delegated (blue) and those who did not (orange). *Punishment* condition, attention-check passers ($n_{\text{deleg}} = 24$, $n_{\text{nodeleg}} = 37$). The only cell with a statistically significant difference is *self decision + good outcome* (top right): delegators expected £0.82 vs non-delegators £0.32 (t -test $p = 0.006$).

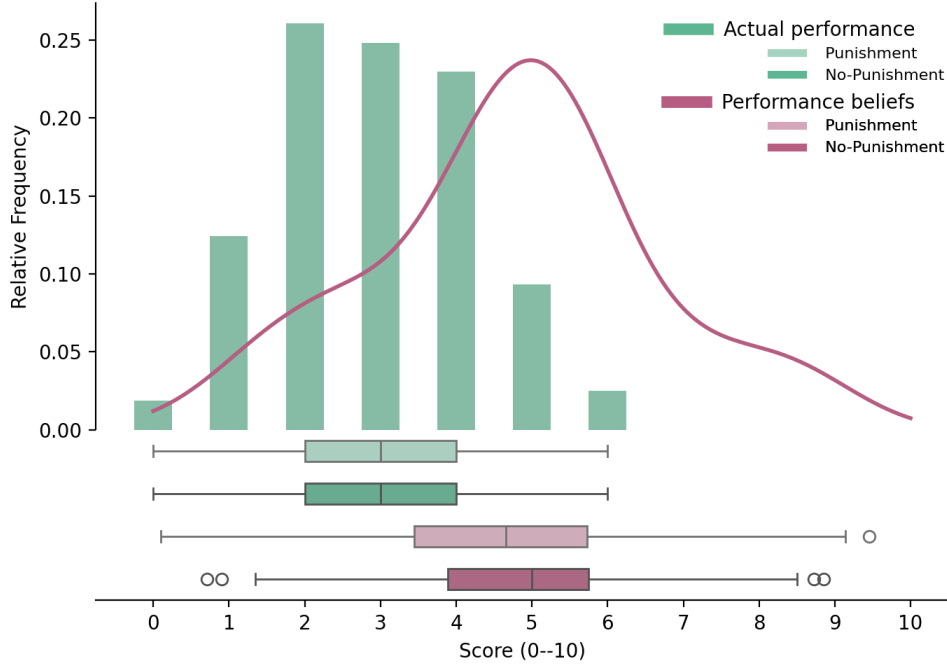
when punishment is on the table. The punishment-belief evidence above shows that Player As do not on average anticipate harsher punishment for delegated decisions, and Player B does not in fact deliver such punishment (Result 2). M1 is therefore inconsistent with both the elicited beliefs and the realized punishment, and we view this candidate as ruled out by the available evidence.

(M2) Effort exertion to outperform the algorithm. An alternative mechanism is that the prospect of punishment motivates Player A to exert greater effort on the final prediction, in the hope of personally producing the high payoff. Although our design controls for relative performance based on the first ten rounds—making the algorithm and Player A equivalent in expectation on the final prediction—potential punishment could in principle shift Player A’s effort margin. The data are inconsistent with the simple form of this story. Among Player As who chose not to delegate, those in the *No-Punishment* condition (where punishment is not possible) outperformed those in the *Punishment* condition: 45.5% of non-delegating Player As in the *No-Punishment* condition earned the high payoff for Player B, compared to 21.7% in the *Punishment* condition. This difference is not driven by ex-ante performance differences across the first ten rounds (*No-Punishment*: 3.12/10; *Punishment*: 2.98/10; two-sided $p = 0.66$); the cross-condition mean score in the first ten rounds is 2.93/10 overall and statistically indistinguishable across conditions (Figure 6), so the M2-relevant gap among non-delegators reflects selection on delegation rather than ex-ante condition differences.¹⁴ Counter-intuitively, increased effort thus appears concentrated among non-delegators in the *No-Punishment* condition, where punishment is *not* on the table—incompatible with a story in which Player A in the *Punishment* condition tries especially hard to outperform the algorithm in order to avoid punishment.

(M3) Process ownership of the outcome. A third mechanism, suggested by the heterogeneity of the delegation decision with respect to Player A’s performance documented in Result 1 above (and Column (3) of Table 1), is that Player As who have a higher subjective probability of producing the high payoff for Player B prefer to bring about that good outcome themselves rather than via the algorithm. The intuition draws on the psychological-ownership literature: people derive utility from being the proximate cause of a good outcome for another, and this utility is asymmetric with respect to outcomes (Steffel et al., 2016). The data are consistent with this mechanism, but the supporting evidence is thin. The negative relationship between performance and delegation in the *Punishment* condition is what process-ownership predicts: better-performing Player As, who are more likely to deliver the high payoff, prefer to

¹⁴Measuring performance change as the difference between the average absolute prediction error across the first ten rounds and the absolute prediction error on the final prediction, we find that Player As in the *No-Punishment* condition improve their performance from rounds 1–10 to round 11, whereas Player As in the *Punishment* condition do not improve. Because all Player As across both conditions faced the same image on the final prediction, this differential improvement cannot be attributed to task-specific variation. We caveat that the comparison is between selected non-delegator subsamples, not between random subsets of all Player As; selection on the very outcome we are measuring is a known limit of this comparison.

Figure 6: Performance in the first ten rounds, by treatment



Note: Distribution of correct predictions (within 10 lbs of the true weight) over the first ten task rounds, by condition. Average performance is 2.93 correct out of ten and is statistically indistinguishable across conditions.

do so personally. The absence of this relationship in the *No-Punishment* condition is consistent with process ownership being amplified by the moral salience that the punishment possibility introduces. We are explicit, however, that the supporting evidence consists of a single heterogeneity pattern in a 60-subject sub-sample and a comparison of selected non-delegator success rates; the pattern is the most consistent reading of the data we have, but not a confirmed mechanism. A direct test would require a treatment that varies process ownership while holding the punishment possibility fixed—for example, an arm in which the algorithm is presented as Player A’s own “tool” rather than as a separate agent.

(M4) Experimenter demand. A final possibility is that the punishment possibility cues Player A to behave more conservatively—for instance, to avoid an action (delegation) that the experimental setting itself implicitly frames as morally questionable. We cannot rule this mechanism out with our data. The most we can say is that two implications of a strong demand story are not borne out: if demand were dominant, Player A’s behavior should track elicited beliefs more tightly than it does, and the sharp performance heterogeneity in the *Punishment* condition (better performers delegating less) would be hard to explain. Neither absence is decisive against the demand reading. We discuss the implications, and the design moves that would identify demand directly, in Section 5.

In sum: M1 (anticipated differential punishment) is inconsistent with the data. M2 (effort to outperform the algorithm) is inconsistent with the data in its simple form. M3 (process ownership) is consistent with the data but supported only by exploratory heterogeneity analysis.

M4 (experimenter demand) cannot be separated from M3 with the present design. The most economical reading—the one we adopt with explicit hedges in the Conclusion—is that the morally charged context introduced by the punishment possibility makes Player A want to be the proximate cause of any good outcome for Player B, while we acknowledge that an experimenter-demand story can produce qualitatively similar patterns. Disentangling the two is the cleanest direction for follow-up work.

5 Discussion and Conclusion

This paper asks two questions about how the prospect of punishment interacts with the use of algorithms as decision aids. Can decision-makers shift responsibility for bad outcomes onto algorithms by delegating their decisions? And does the anticipation of being held responsible motivate delegation in the first place? We address both questions in a preregistered online experiment in which Player A makes a payoff-relevant prediction for Player B and may delegate the prediction to an algorithm calibrated to perform as well as Player A herself. A between-subject manipulation switches off punishment by Player B in the *No-Punishment* condition; the *Punishment* preserves the natural setting in which the prospect of punishment is on the table.

Two findings emerge. First, and most striking, the *prospect* of punishment *reduces*, rather than increases, the propensity to delegate to the algorithm: 42.5% of Player As delegate when punishment is possible, compared to 59.3% when it is not. This direction is the opposite of what the responsibility-avoidance literature would predict, and we view the reversal as the paper’s main empirical contribution—noting that, because the result reverses the preregistered direction, the analyses we use to interpret it (Section 4) are exploratory rather than confirmatory. Second, conditional on the outcome for Player B, we cannot reject the hypothesis that delegated and self-made decisions are punished equally. The point estimate runs in the predicted direction (lower punishment under delegation in the bad-outcome cell) but is small relative to the test’s confidence interval at the achieved sample size; we therefore frame Result 2 as a failure to detect insulation rather than as positive evidence that delegation cannot insulate. Player A does not appear to be able to use this particular algorithm as a meaningful blame shield in the bounded sense the test can assess.

Mechanism. Why does punishment reduce, rather than increase, delegation? The data are inconsistent with a simple anticipation mechanism: Player As do not on average expect harsher punishment for delegated decisions, and they correctly anticipate the absence of differential punishment that we observe in Player B’s actual choices. The data are likewise inconsistent with the simple form of an effort story in which the prospect of punishment motivates Player A to outperform the algorithm: among non-delegators, those in the *No-Punishment* condition (where punishment is impossible) outperform those in the *Punishment* condition, the reverse

of what such a story implies. The reading most consistent with the data is one of *process ownership*: Player As who actively choose not to delegate take responsibility for the outcome, exert correspondingly higher effort, and that share is higher when punishment is on the table. The supporting evidence is exploratory rather than confirmatory, however, and rests on a single performance-by-delegation heterogeneity pattern in a 60-subject sub-sample (Column (3) of Table 1); we cannot, with this design, separate process ownership cleanly from an experimenter-demand reading in which the punishment cue itself nudges Player A toward direct engagement. We therefore present process ownership as the most consistent reading of the data we have, not as a confirmed mechanism. Disentangling it from demand is a clean direction for follow-up work, sketched below.

This refinement is consistent with, but goes beyond, the standard reading of the responsibility-avoidance literature. The classic finding—that delegation reduces punishment of the principal and that this prospect motivates delegation—was established in dictator-game settings where the underlying choice is between a known fair and a known unfair allocation, the intermediary is human, and the principal can be held accountable specifically for choosing a particular intermediary (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013). When the intermediary is an algorithm calibrated to mimic the principal and the underlying decision involves genuine uncertainty, none of the standard mechanisms through which delegation can shift responsibility—signal value of the delegation choice, the human-human punishment channel, the certainty of the unfair outcome—are available. What remains is a baseline outcome-based punishment, supplemented by an asymmetric desire to be the proximate cause of good outcomes when the moral stakes are salient.

Implications, with explicit bounds. Our findings speak to the design of accountability regimes governing algorithmic decision-making. The European Union’s AI Act (European Union, 2024) and analogous regulatory frameworks elsewhere place increasing emphasis on assigning responsibility to human users of high-risk algorithmic systems. The implicit theory of behavior underlying these provisions is that holding humans accountable for algorithm-mediated decisions *disciplines* their use of those algorithms. Our results offer a behavioral complement to this view: when punishment is on the table, decision-makers in our setting do not simply delegate-and-blame the algorithm; they choose to take direct responsibility for the decision more often. From the perspective of efficiency, this need not be welfare-increasing: in domains where algorithms genuinely outperform humans, holding humans liable for outcomes may push decision-makers away from the more accurate technology. The trade-off between encouraging algorithmic adoption and assigning meaningful responsibility is, on this reading, sharper than the existing literature has suggested.

We note explicitly the bounds of this reading. Our evidence is a single online experiment with 322 UK Prolific participants on a stylised weight-prediction task, with stakes of £5 for Player B and a punishment range of £0–£2. The high-stakes algorithmic domains that motivate the regulatory debate—medical diagnosis, judicial risk assessment, hiring, credit allocation—differ

from our setting in stake magnitude, professional context, expert training, the institutional structure of accountability, and the kind of population at the decision margin. We therefore view the result as identifying a behavioral channel that exists in principle and is detectable in our setting, not as a calibration of how big a lever accountability is in any specific applied domain. Mapping the magnitude of the effect against external benchmarks—e.g., physician adoption of diagnostic algorithms under malpractice exposure, or loan-officer deviation from credit-scoring algorithms under fair-lending audit—would require either domain-specific replication or instruments to which we do not have access. We flag this gap explicitly so that readers can calibrate the policy implication appropriately.

Limitations. Several caveats are in order. The belief measure we elicit is unincentivised and noisy; it conveys ordinal patterns reliably but cannot support fine-grained quantitative claims about Player A’s anticipations. Two of the four belief scenarios are hypothetical conditional on the delegation decision, and all four are hypothetical in the *No-Punishment* condition; this further limits the inference, and we plan to report the H2 patterns separately for the payoff-relevant and the hypothetical cells once the analysis pipeline is in place. We use the strategy method for Player B’s punishment decision, which is standard in this literature but may yield different absolute levels of punishment than a direct elicitation. Player A also makes the final prediction whether or not she ultimately decides to delegate, which equates the time cost of the two options but may signal her commitment differently than a design in which the prediction is skipped under delegation. We did not include a separate manipulation-comprehension check verifying that Player A internalised the punishment-on/off distinction; we relied on the prominent screen-level framing and on attention-check pass rates as indirect evidence, but a future replication should add this check. The absolute stakes are modest (£5 high payoff, £0–2 punishment range): we cannot speak to how the effect scales with stakes, and a stake-magnitude robustness check is the first item in the future-research list below. Our sample is drawn from UK Prolific participants, which limits external validity along the dimensions in which Prolific samples differ from broader populations and from professional decision-makers in the high-stakes algorithmic domains we discuss above.

A natural concern is whether the punishment possibility cues Player A to behave more conservatively for reasons unrelated to the substantive question—an experimenter-demand effect. We cannot rule this concern out from the data. The direction of any such effect is also ambiguous: a decision-maker who is reminded of being judged could plausibly delegate *more* (to shed responsibility) or *less* (to demonstrate effort and commitment); the literature does not unambiguously favor one over the other. We note two patterns weakly inconsistent with a strong demand reading—behavior does not track elicited beliefs as tightly as a pure-demand story would predict, and the sharp performance heterogeneity in the *Punishment* condition (better performers delegating less) is harder to motivate from demand alone—but we treat these as suggestive rather than dispositive. A direct test in the spirit of [de Quidt et al. \(2018\)](#) would be the natural next step.

Directions for future research. Several extensions would strengthen the inference. A direct-method companion would test the robustness of Result 2 against the strategy-method elicitation, and a meaningfully larger sample would let us put a tight confidence interval on the small but persistent point estimate of insulation under delegation. A condition in which Player A is told explicitly that Player B is unaware of the equal-performance calibration would test whether the relative-performance signal is the binding margin distinguishing our results from [Feier et al. \(2022\)](#). Conditions that experimentally vary the algorithm’s performance relative to Player A (above, equal, below) would map the generality of the process-ownership reading. A demand-effect probe along the lines of [de Quidt et al. \(2018\)](#) would put a credible upper bound on the contribution of M4. A stake-magnitude robustness check, varying the £5/£1 high/low payoffs and the £0–£2 punishment range, would speak to whether the effect scales with stakes in the direction that policy-relevant settings would imply. Finally, replication in a different country, in a different decision domain (e.g., a medical-triage or hiring vignette), or with a sample of professional decision-makers rather than online crowd-workers would speak directly to the external-validity bounds we describe above.

Taken together, our results suggest a more textured story than the one that has emerged from the existing literature on responsibility avoidance and algorithm aversion. Within the bounds of the setting we study, the capacity to use algorithms as a tool for shifting responsibility appears limited, and the prospect of being held responsible—rather than encouraging delegation as a hedge—appears to push decision-makers towards direct engagement with the consequential choice. Whether the same channel operates in the high-stakes algorithmic domains that animate the regulatory debate is an empirical question that this study can frame but cannot, on its own, answer.

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Appendix

A Additional Tables

A.1 Condition balance — Player A

Table 2: Condition balance — Player A

Variable	<i>Punishment</i>	<i>No-Punishment</i>	<i>p-value</i>
Age	39.026	37.877	0.552
Female	0.487	0.494	1.000
Socio-economic status	5.179	5.321	0.564
Went to uni	0.575	0.556	0.928
Technology score	2.087	2.284	0.322
Leadership position	0.362	0.333	0.824

Note: Mean values by condition for Player A, with *p*-values from two-sided *t*-tests for continuous variables and Chi-squared tests for binary variables. *Socio-economic status* is self-reported on a 1–10 scale; *Went to uni* indicates a university degree; *Technology score* is constructed from weekly device usage, programming skills, cryptocurrency knowledge, technology use at work, and similar items; *Leadership position* indicates self-reported management or supervisory experience.

We do not separately report a balance table for Player B. Player B’s punishment behavior is identified within-subject across the four scenarios of the strategy method, so balance across conditions is not strictly required for the punishment analysis.

A.2 Full regression results — delegation decision

A.3 Robustness — delegation decision

A.4 Determinants of Player B’s chosen punishment

A.5 Robustness — Player B’s chosen punishment

B Bonus computation

Player A spent on average 12 minutes on the experiment and earned an average bonus of £2.49 on top of the Prolific show-up fee; Player B spent on average 8 minutes and earned an average bonus of £2.30. Bonuses are computed deterministically from the cleaned data per the experimental rules: random-round task payoff (expectation over the uniform draw across rounds 1–10); belief-about-own-score payoff (£0.30 times the probability mass placed on the realized score); belief-about-distribution payoff (expectation over the uniform random draw of which score is checked, paying £0.30 if the estimate is within 5 percentage points of the true population share); and—for Player A in the *Punishment* condition—expected punishment

Table 3: Determinants of the delegation decision — full controls

	Dependent variable: <i>Delegation</i>		
	<i>Full sample</i>		<i>Punishment only</i>
	(1)	(2)	(3)
No-Punishment indicator	0.677 (0.320)	0.757 (0.334)	
Punishment beliefs: bad outcome			0.937 (0.751)
Punishment beliefs: good outcome			1.579 (0.741)
Task performance		−0.192 (0.128)	−0.609 (0.236)
Age		−0.004 (0.015)	0.064 (0.030)
Female		0.350 (0.369)	1.334 (0.793)
Socio-economic status		−0.058 (0.116)	−0.003 (0.219)
Went to uni		0.091 (0.364)	−0.433 (0.707)
Technology score		−0.078 (0.153)	0.059 (0.283)
Leadership position		0.676 (0.367)	−0.387 (0.729)
Constant	−0.302 (0.226)	0.402 (0.930)	−1.842 (2.103)
N	161	159	60
Pseudo R^2	0.020	0.049	0.246

Note: Logit estimates with robust (HC1) standard errors in parentheses. Two subjects in Column (2) are dropped due to missing socio-demographic data. Column (3) restricts to the *Punishment* condition and excludes subjects who failed the attention check on the belief-elicitation screen, as preregistered. The belief variables in Column (3) are within-subject differences in expected punishment between non-delegated and delegated decisions for each outcome. Following AEA editorial policy, significance stars are not reported in the table.

Table 4: Determinants of the delegation decision — robustness

	Dependent variable: <i>Delegation</i>		
	<i>Full sample</i>		<i>Punishment</i> (no excl.)
	(1)	(2)	(3)
No-Punishment indicator	0.771 (0.336)	0.677 (0.320)	
Task performance	−0.277 (0.149)		−0.395 (0.185)
Overconfidence	−0.103 (0.094)		
Punishment beliefs: bad outcome (no del. – del.)			−0.277 (0.499)
Punishment beliefs: good outcome (no del. – del.)			0.812 (0.529)
Controls	yes	no	yes
Attention-check excl.	yes (Player A)	n/a	no
Sample	Full	Full	Punishment
N	159	161	78
Pseudo R^2	0.055	0.020	0.114

Note: Robustness specifications for the delegation logit. Column (1) adds *overconfidence* (defined as the weighted-average belief about own performance minus actual performance) to the main full-sample specification. The treatment effect remains: *No-Punishment indicator* $p = 0.022$. *Overconfidence* is not significantly associated with delegation $p = 0.272$. Column (2) reproduces the unconditional treatment effect for completeness ($p = 0.034$; identical to the main Col (1)). Column (3) restricts to the *Punishment* condition and includes the within-subject belief differences but *does not* exclude subjects who failed the belief-screen attention check; the heterogeneity result is preserved: *Task performance* $p = 0.033$; *Punishment beliefs: good outcome* $p = 0.125$. Robust HC1 standard errors in parentheses. Following AEA editorial policy, significance stars are not reported in the table.

Table 5: Determinants of Player B's chosen punishment

Dependent variable: <i>Punishment</i>	
	<i>Punishment</i> (£)
	(1)
Delegated	0.001 (0.054)
Bad outcome	0.260 (0.079)
Delegated $\times$ Bad outcome	-0.078 (0.052)
Age	-0.001 (0.006)
Female	-0.065 (0.143)
Socio-economic status	0.072 (0.049)
Went to uni	-0.099 (0.136)
Technology score	0.030 (0.058)
Leadership position	0.060 (0.163)
Constant	0.027 (0.344)
N (subject-by-cell)	268
Adj. R^2	0.034

Note: OLS estimates of Player B's chosen punishment on a delegation indicator, a bad-outcome indicator, their interaction, and socio-demographic controls. Standard errors clustered at the Player B level (in parentheses). Sample: Player B subjects in the *Punishment* condition who passed the punishment-elicitation attention check, expanded over the four (delegation, outcome) cells of the strategy method. Two-sided p -values: *Delegated* $p = 0.989$; *Bad outcome* $p = 0.001$; *Delegated* $\times$ *Bad outcome* $p = 0.137$. Following AEA editorial policy, significance stars are not reported.

Table 6: Determinants of Player B’s chosen punishment — robustness

	Dependent variable: <i>Punishment</i>	
	<i>Excl.</i>	<i>Incl. att. failers</i>
	(1)	(2)
Delegated	0.001 (0.054)	0.032 (0.052)
Bad outcome	0.260 (0.080)	0.256 (0.071)
Delegated $\times$ Bad outcome	−0.078 (0.052)	−0.103 (0.058)
Player B belief about Player A perf. (wa)	0.001 (0.037)	−0.020 (0.032)
Age	−0.001 (0.006)	0.002 (0.005)
Female	−0.064 (0.140)	−0.100 (0.128)
Socio-economic status	0.072 (0.049)	0.083 (0.047)
Went to uni	−0.098 (0.142)	−0.196 (0.140)
Technology score	0.030 (0.059)	0.073 (0.053)
Leadership position	0.061 (0.164)	0.083 (0.142)
Constant	0.020 (0.374)	−0.077 (0.328)
Attention-check excl.	yes	no
N (subject-by-cell)	268	320
Adj. R^2	0.030	0.077

Note: Punishment regression with Player B’s belief about Player A’s performance (*wa_difficulty*) added as a regressor. Column (1) excludes Player Bs who failed the punishment-screen attention check (preregistered); Column (2) keeps them. Standard errors clustered at the Player B level. The delegation, outcome, and interaction coefficients are stable across both specifications, and the belief regressor is not significantly associated with chosen punishment. H2 paired-test on the bad-outcome cell: with exclusion paired- t $p = 0.153$, Wilcoxon $p = 0.533$ ($n = 67$); without exclusion paired- t $p = 0.131$, Wilcoxon $p = 0.544$ ($n = 80$). Following AEA editorial policy, significance stars are not reported.

from the matched Player B’s strategy-method choice in the realized cell, with £0.10 per cell of strategy-method punishment cost for Player B. The reported bonuses exclude the risk-elicitation MPL payoff (the raw-data column was not retained in the cleaned files); including it would add roughly £0.30–£0.50 per subject.

C Ex-ante power calculations

The cell size of 80 subjects per role per condition was set to detect, at conventional power, the effect sizes anticipated from the responsibility-avoidance literature. We report the minimum detectable effect (MDE) for each preregistered test below, computed at $\alpha = 0.05$ (two-sided) and power = 0.80. For H1, the relevant test is a two-sample z -test of proportions (delegation rate in *Punishment* vs *No-Punishment*); the MDE is approximately invariant to the assumed baseline rate. For H2, the relevant test is a paired-mean-difference test on Player B’s punishment in the bad-outcome cell; because the within-subject standard deviation σ_d of the punishment difference is unknown ex ante, we report the MDE for a range of values of σ_d expressed as multiples of the equivalence bound used in Result 2 of the body of the paper ($\delta_a = £0.20$, equal to 10% of the maximum punishment of £2).

Table 7: Minimum detectable effects under the preregistered design

Hypothesis	Assumption	Minimum detectable effect
H1 (delegation, two-proportion test)	Baseline $p_0 = 0.30$	$\Delta = 0.216$ (21.6 pp)
H1 (delegation, two-proportion test)	Baseline $p_0 = 0.50$	$\Delta = 0.215$ (21.5 pp)
H1 (delegation, two-proportion test)	Baseline $p_0 = 0.70$	$\Delta = 0.179$ (17.9 pp)
H2 (paired, within-subject)	$\sigma_d = 0.5 \delta_a = £0.10$	$\Delta = £0.031$
H2 (paired, within-subject)	$\sigma_d = 1 \delta_a = £0.20$	$\Delta = £0.063$
H2 (paired, within-subject)	$\sigma_d = 2 \delta_a = £0.40$	$\Delta = £0.125$
H2 (paired, within-subject)	$\sigma_d = 3 \delta_a = £0.60$	$\Delta = £0.188$
H2 (paired, within-subject)	$\sigma_d = 4 \delta_a = £0.80$	$\Delta = £0.251$

Note: Both tests assume $n = 80$ per role per condition, $\alpha = 0.05$ two-sided, power = 0.80. For H2, σ_d is the within-subject standard deviation of the punishment difference (delegated minus non-delegated, conditional on bad outcome) and is expressed as a multiple of the anchor effect $\delta_a = £0.20$ (the equivalence bound used in Result 2 in the body of the paper, equal to 10% of the maximum punishment of £2). Δ is the corresponding minimum detectable effect in pounds.

D Preregistration and deviations

The experiment was preregistered at AsPredicted #133980, available at aspredicted.org/hs9az8.pdf and registered on 30 May 2023 prior to the main wave of data collection. The preregistration commits to the two research questions stated in the body as H1 and H2, the sample-size target (320 subjects total, 80 per role per condition), the desktop-only restriction, the use of the strategy method to elicit Player B’s punishment, and a set of primary statistical tests (parametric

t -tests and non-parametric MWU and Kolmogorov–Smirnov tests for both comparisons; probit regressions for the delegation decision and linear regressions for punishment behavior with treatment, beliefs, and socio-demographics as predictors).

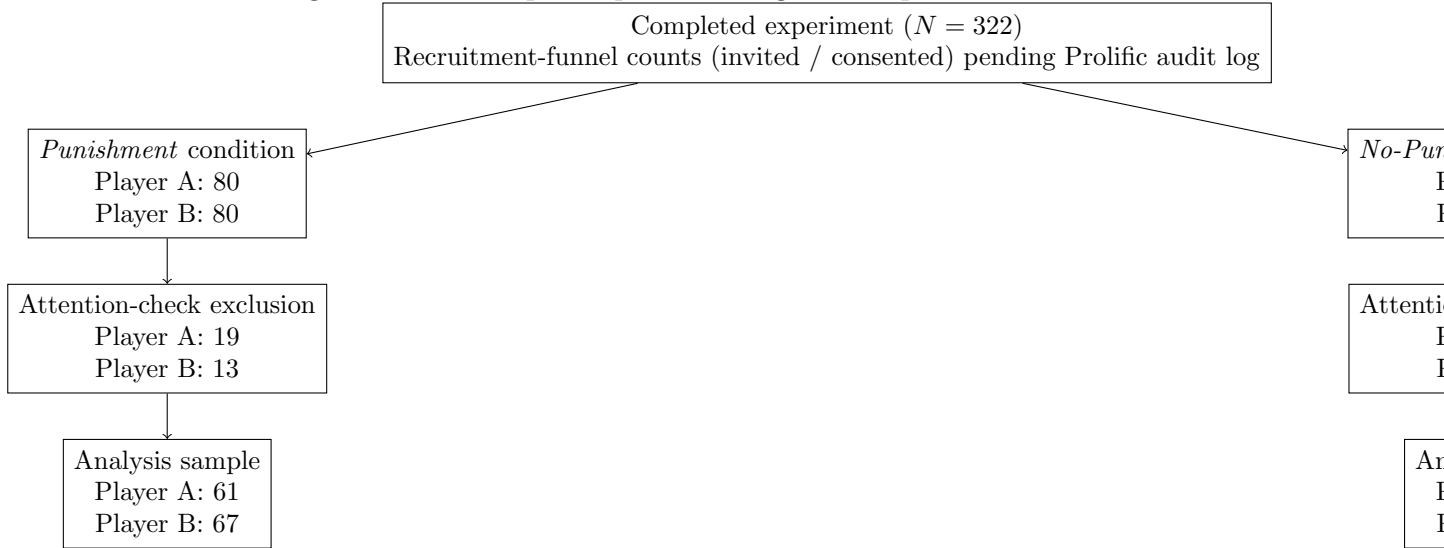
The preregistration also notes that a pilot of 25 observations had been collected in the *Punishment* treatment prior to registration; these observations are pooled with the main wave in all analyses, as preregistered. Two secondary analyses are pre-stated in the registry: the role of own-performance and beliefs about own performance (developed as M2 in Section 4), and the consistency of Player A’s beliefs about expected punishment with her delegation decision (developed as M3 in Section 4).

The table below summarizes every analysis reported in the body of the paper, the corresponding preregistered specification, and any deviation.

E Flow of participants

The flow of participants through the experiment, by condition and role, is shown in Figure 7. Recruitment-funnel counts (invited / consented) are pending the Prolific audit log.

Figure 7: Flow of participants through the experiment



F Additional Figures

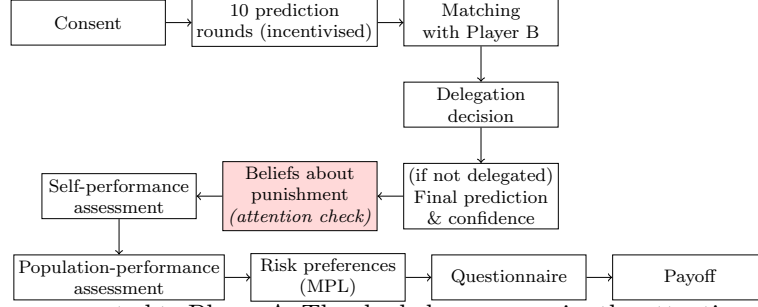
F.1 Experimental timelines

F.2 Punishment frequencies

Table 8: Preregistered tests, reported tests, and deviations

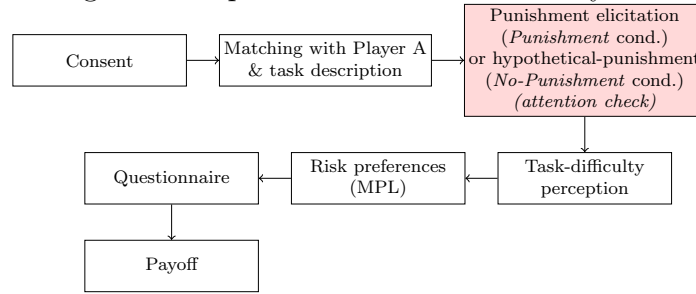
Preregistered specification	As reported	Deviation / rationale
H1: between-treatment test of delegation rates (<i>Punishment</i> vs <i>No-Punishment</i>); preregistered tests are parametric <i>t</i> -test, MWU, KS. Direction not specified in the registry.	Pearson Chi-squared and Fisher’s exact, two-sided ($p = 0.033$ / $p = 0.041$).	Test family substituted (Chi-squared and Fisher’s exact are the standard tests for a binary outcome; the preregistered <i>t</i> -test, MWU and KS tests on a binary outcome are equivalent to the proportion test). The realized direction is opposite to that anticipated by the responsibility-avoidance literature on which we drew.
H2: paired test of Player B punishment in the bad-outcome cell, delegated vs non-delegated, attention-check passers only	Paired <i>t</i> -test ($p = 0.153$); Wilcoxon signed-rank ($p = 0.533$).	As preregistered.
Probit / logit of delegation on condition + controls (Table 1, Cols (1)–(2))	Logit, as reported.	As preregistered. (Probit and logit are equivalent for our purposes; we report logit for ease of marginal-effect interpretation.)
Logit on <i>Punishment</i> subsample with belief differences (Col (3))	As reported.	Specification preregistered; sample restriction follows the preregistered attention-check rule.
Pre-stated secondary analysis: own-performance and beliefs about own performance	Reported as M2 in Section 4.	As preregistered (secondary).
Pre-stated secondary analysis: beliefs about expected punishment vs delegation decision	Reported as M3 in Section 4.	As preregistered (secondary).
Mechanism analyses M1 (process ownership) and M4 (experimenter demand) (Section 4)	As reported.	<i>Not preregistered.</i> Treated as exploratory.

Figure 8: Experimental timeline — Player A



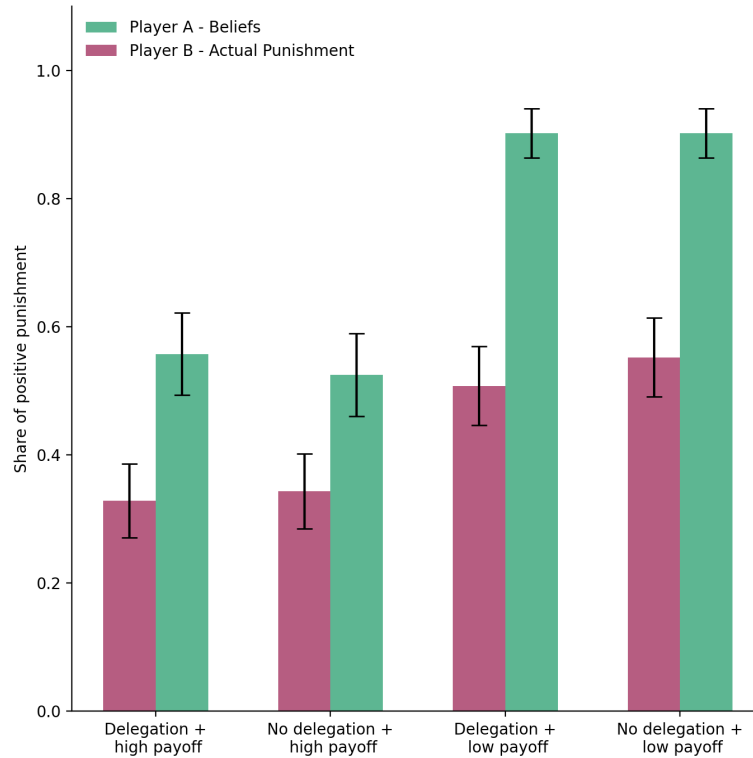
Note: Screen sequence presented to Player A. The shaded screen carries the attention check used to identify subjects excluded from belief-related analyses. The final-prediction-and-confidence screen is skipped if Player A delegates to the algorithm.

Figure 9: Experimental timeline — Player B



Note: Screen sequence presented to Player B. The shaded screen carries the attention check used to identify subjects excluded from punishment-related analyses. In the *No-Punishment* condition, no actual punishment is implemented; the elicitation is hypothetical and serves to maintain comparability of the screen sequence across conditions.

Figure 10: Frequencies of punishment and beliefs about punishment



Note: Share of Player Bs choosing strictly positive punishment in each of the four scenarios (solid bars) and share of Player As assigning a strictly positive expected punishment in each scenario (hollow bars).